

Network Latency in Distributed LLM Inference

PROGRESS UPDATE: MAY 13, 2026

Kelsey Penners

Big Refactor of Code & Data

- Restructured results directory to follow a more standardized hierarchy:
  - `{experiment}/{gpu_type}/tp{n}/{interconnect}/{timestamp}/`
- Measuring GPU power usage for every run
- CLI tool for running experiments: flags and JSON config files allow for easier model, TP, interconnect configuring
- Updated parsing and plotting code so works with new structure and is much easier to add new configs!

Parameter	Purpose
`--experiment`	Benchmark config to run
`--model`	Model used for run
`--gpu-ids`	CUDA GPU IDs for run
`--interconnect`	NCCL communication mode
`--nccl-debug`	Enable NCCL logging
`--tp`	TP degree
`--hardware`	GPU config for logging

funnotch Hardware: Heterogeneity

```
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 580.159.03             Driver Version: 580.159.03     CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla V100-PCIE-32GB           Off |   00000000:61:00.0 Off |                    0 |
| N/A   46C    P0             99W /  250W |   29458MiB /  32768MiB |     87%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   1  Tesla V100S-PCIE-32GB          Off |   00000000:86:00.0 Off |                    0 |
| N/A   65C    P0            157W /  250W |   29458MiB /  32768MiB |     86%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   2  Tesla V100S-PCIE-32GB          Off |   00000000:DA:00.0 Off |                    0 |
| N/A   60C    P0            141W /  250W |   29470MiB /  32768MiB |     86%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
|   3  Tesla V100-PCIE-32GB           Off |   00000000:DB:00.0 Off |                    0 |
| N/A   56C    P0            113W /  250W |   29518MiB /  32768MiB |     86%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
```

- `funnotch` has mixed GPUs:
  - 2×V100 (GPU0,3), 2×V100S (GPU1,2)
- Asymmetric PCIe topology: only GPU2 and 3 share PIX connection
- P2P access unavailable between other GPUs, affecting TP=4 communication

vLLM Warning:

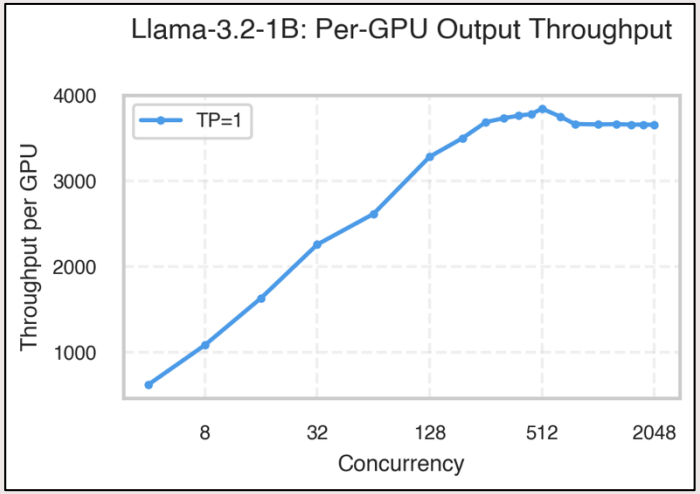
```
WARNING 05-10 20:21:23 [cuda.py:608] Detected different devices in the system: Tesla V100-PCIE-32GB, Tesla V100S-PCIE-32GB, Tesla V100S-PCIE-32GB, Tesla V100-PCIE-32GB. Please make sure to set `CUDA_DEVICE_ORDER=PCI_BUS_ID` to avoid unexpected behavior.
```

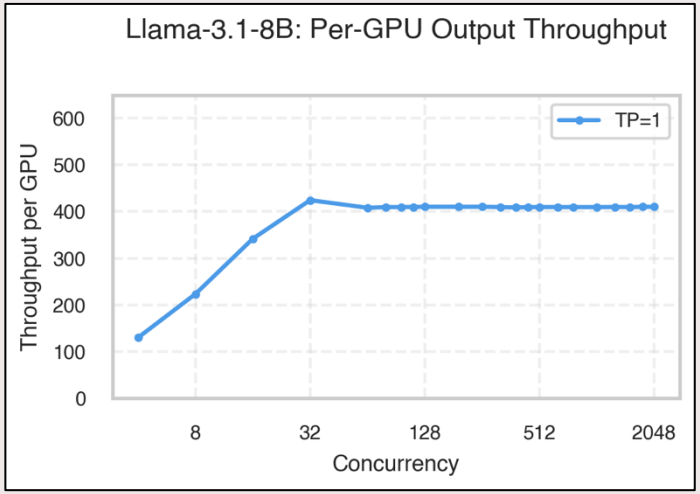
```
      GPU0  GPU1  GPU2  GPU3  CPU Affinity    NUMA Affinity  GPU NUMA ID
GPU0   X     SYS   SYS   SYS   0-23,48-71      0              N/A
GPU1   SYS    X    NODE  NODE  24-47,72-95     1              N/A
GPU2   SYS   NODE   X    PIX   24-47,72-95     1              N/A
GPU3   SYS   NODE  PIX    X    24-47,72-95     1              N/A

Legend:

  X    = Self
  SYS  = Connection traversing PCIe as well as the SMP interconnect between NUMA nodes (e.g., QPI/UPI)
  NODE = Connection traversing PCIe as well as the interconnect between PCIe Host Bridges within a NUMA node
  PHB  = Connection traversing PCIe as well as a PCIe Host Bridge (typically the CPU)
  PXB  = Connection traversing multiple PCIe bridges (without traversing the PCIe Host Bridge)
  PIX  = Connection traversing at most a single PCIe bridge
  NV#  = Connection traversing a bonded set of # NVLinks
```

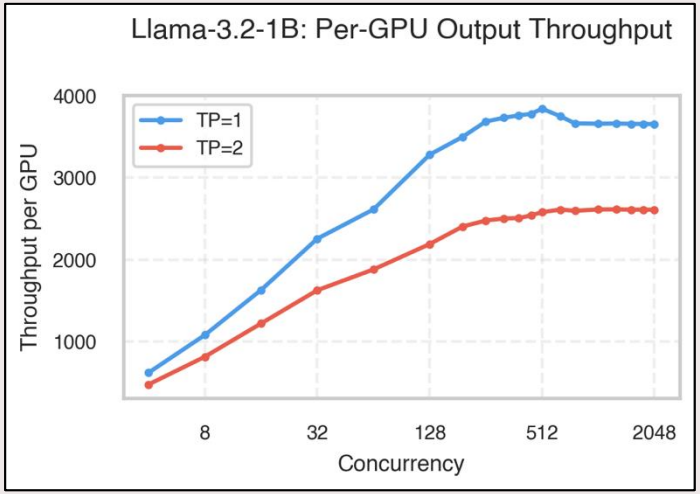
Scaling with Models of Different Sizes: `funnotch`

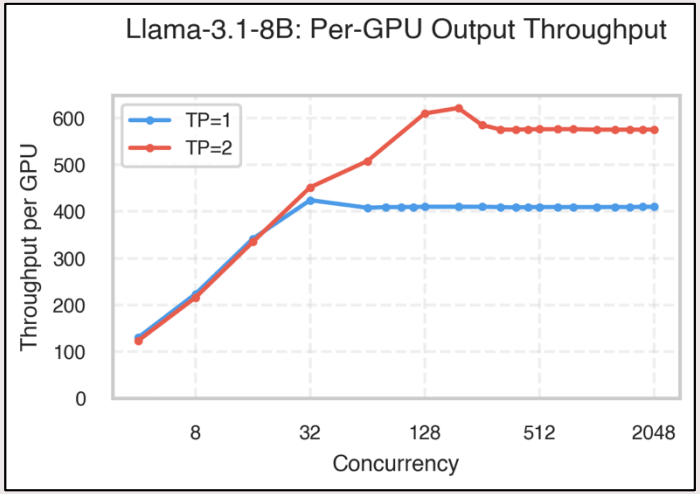




- Smaller models (1B) shows decreasing per-GPU efficiency as TP increases
- For the 8B model, TP=2 exceeds TP=1 per-GPU throughput: increased KV cache offsets communication cost
- As TP degree increases, communication overhead starts to outweigh memory capacity benefits

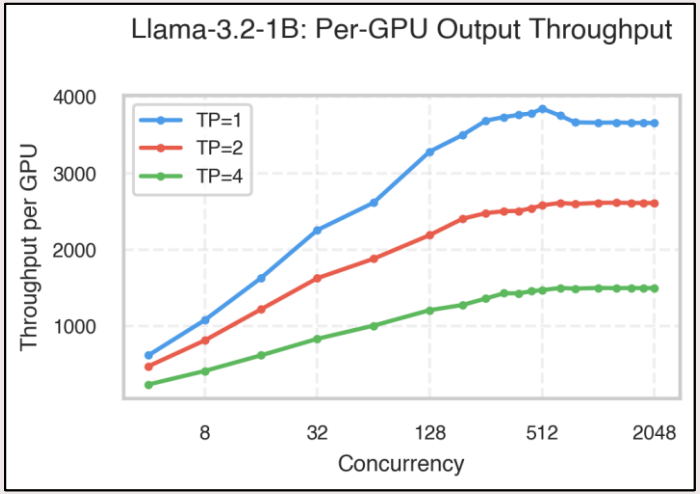
Scaling with Models of Different Sizes: `funnotch`

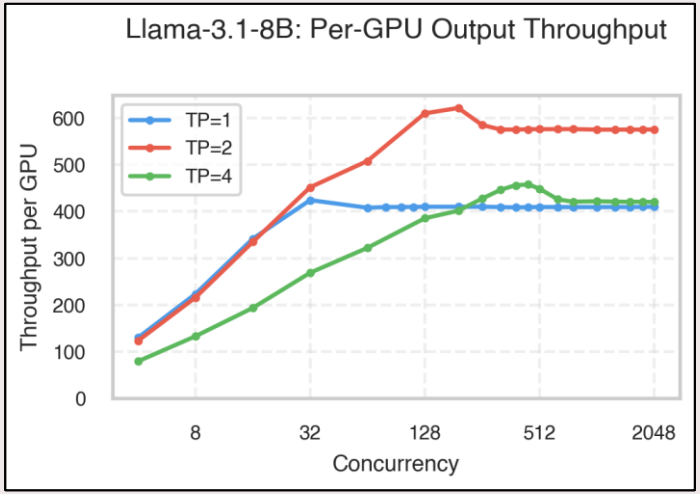




- Smaller models (1B) shows decreasing per-GPU efficiency as TP increases
- For the 8B model, TP=2 exceeds TP=1 per-GPU throughput: increased KV cache offsets communication cost
- As TP degree increases, communication overhead starts to outweigh memory capacity benefits

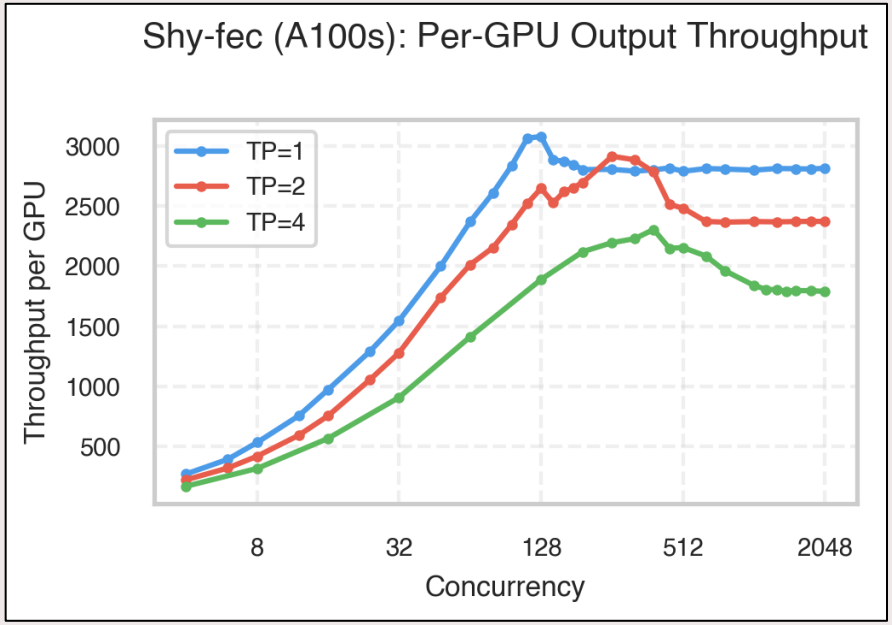
Scaling with Models of Different Sizes: `funnotch`





- Smaller models (1B) shows decreasing per-GPU efficiency as TP increases
- For the 8B model, TP=2 exceeds TP=1 per-GPU throughput: increased KV cache offsets communication cost
- As TP degree increases, communication overhead starts to outweigh memory capacity benefits

Recap: Llama-3.1-8B on shy-fec



Next Steps

- Run TP scaling experiments with LLMs of varying sizes on both servers (Llama 3.2-**1B** and Llama 2-**13B**)
  - To how observe how model size affect scaling efficiency
- Begin writing TP scaling results and interconnect comparison sections
- Also run interconnect experiments with TP=4?
- Look into `funnotch` heterogeneity more closely?
- Power usage investigation?

Discussion

- Citation style in thesis: how should I cite when paraphrasing versus referencing results?

Thank you!